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Machine Learning Approach to Car Model Identification via Images

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Abstract

This project explores the application of machine learning techniques [9], [10] for car model classification using two distinct neural network models: ConvLSTM2D [2] and VGG16 [1]. ConvLSTM2D, which integrates convolutional layers with LSTM units, is designed to handle spatiotemporal data. VGG16, a pre-trained model on the ImageNet [5] dataset, is fine-tuned specifically for car model identification. We utilized a diverse dataset of car images, performing comprehensive preprocessing and standardization to prepare the data for training and evaluation. Experimental results demonstrate the comparative performance of ConvLSTM2D and VGG16, comparing their strengths and limitations through accuracy metrics of training. Our findings suggest that while ConvLSTM2D can handle sequential image data, VGG16 performs better on static car model classification tasks.

1. INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

In today's technologically advanced world, the application of deep learning in computer vision has become increasingly complex. The automotive industry is one area where these advancements have shown immense potential. Automated vehicle identification and classification systems are becoming critical components in various applications, ranging from traffic management systems and autonomous vehicles to smart parking solutions and enhanced security protocols. Accurate and efficient car model recognition systems can significantly contribute to these applications, offering improvements in operational efficiency and accuracy. The rapid development of deep learning techniques has provided powerful tools to tackle complex image classification tasks.

Car model recognition systems have diverse applications across various domains. In traffic management, they can be used to monitor and analyze traffic flow, identify congested areas, enforce traffic regulations by recognizing specific car models involved in violations, detect frauds in online e-commerce, and many others. In autonomous driving, accurate car model identification is crucial for understanding the surrounding environment, enabling the vehicle to make informed decisions based on the presence and behavior of different car models. In security and surveillance, car model recognition can aid in tracking stolen vehicles, monitoring suspicious activities, and enhancing overall situational awareness. Furthermore, in commercial applications such as smart parking and toll collection, these systems can streamline operations by automatically identifying and categorizing vehicles, leading to improved efficiency and user experience.

1.2 PROBLEM STATEMENT

Despite advancements in image processing and machine learning, accurately identifying car models from images remains a challenging task. Traditional methods, such as feature-based or handcrafted algorithms, often struggle with variations in lighting, angle, occlusion, and background noise.

While deep learning models have shown promise, there is still a need for a more efficient and scalable approach that can handle real-world conditions effectively.

1.3 OBJECTIVES

The main objectives of this project are:

- To develop a machine learning-based system for car model identification using image processing techniques.
- To leverage deep learning architectures, specifically Convolutional Neural Networks (CNNs), for accurate car model classification.
- To implement and evaluate ConvLSTM2D and VGG16 models for their effectiveness in recognizing different car models.
- To analyze the impact of various factors such as lighting conditions, occlusions, and different camera angles on model performance.
- To explore real-world applications of car model recognition, including traffic management, autonomous driving, security, and commercial use cases.

1.4 SCOPE AND LIMITATIONS

Despite significant advancements, several challenges remain in developing robust car model recognition systems. The diversity of car models, variations in lighting conditions, occlusions, and viewpoint changes are some of the factors that can affect the accuracy and reliability of these systems. Therefore, leveraging pre-trained models like ConvLSTM2D and VGG16, which have been trained on large-scale datasets and have demonstrated strong generalization capabilities, can provide a solid foundation for building effective car model classifiers

2. LITERATURE REVIEW / RELATED WORK

2.1 EXISTING STUDIES

A. Deep Learning for Image Classification

Deep learning has revolutionized image classification tasks by leveraging complex neural network architectures that are capable of learning intricate patterns and features directly from raw pixel data. Traditional Convolutional Neural Networks (CNNs) such as VGG16, ResNet, and Inception have demonstrated exceptional performance in various computer vision tasks. These architectures employ stacked convolutional layers followed by fully connected layers to extract hierarchical representations of input images [1], [3], [4]. Recent advancements have introduced hybrid architectures like ConvLSTM2D, which integrate convolutional layers with Long Short-Term Memory (LSTM) units. This fusion enables models to not only capture spatial features but also learn temporal dependencies in sequential data. Such innovations have proven crucial in applications where understanding both spatial structures and temporal dynamics is essential [2].

B. VGG16

VGG16, proposed by Simonyan and Zisserman [1], is known for its simplicity and effectiveness in image classification tasks. With a standardized architecture consisting of 16 layers, including multiple convolutional layers followed by max-pooling and fully connected layers [11], VGG16 achieves high accuracy by progressively learning abstract features from raw pixels. Its straightforward design and strong empirical performance make it a popular choice for benchmarking and transfer learning in computer vision.

C. ConvLSTM2D

ConvLSTM2D, introduced by Xingjian et al. [2], extends the capabilities of traditional LSTM networks by integrating them with convolutional operations. This architecture is particularly suited for tasks requiring spatiotemporal modeling, such as video analysis and weather forecasting. By embedding convolutional layers within the LSTM

framework, ConvL STM2D preserves spatial information across time, enabling robust predictions in dynamic sequences.
Recent research efforts have focused on enhancing the efficiency and scalability of ConvLSTM2D models, exploring novel architectures and optimization strategies to address computational challenges while maintaining predictive accuracy.

2.2 COMPARISON WITH EXISTING WORK

This project builds upon previous studies by combining the advantages of **CNN-based classification** and **ConvLSTM2D** to improve the robustness of car model recognition under real-world conditions. Unlike prior studies that focus solely on static image classification, this study aims to address:

- **Real-world challenges** such as varying illumination, occlusions, and different viewpoints, have been limitations in previous works.
- **The use of ConvLSTM2D**, which allows capturing temporal dependencies if multiple frames are available, enhances recognition accuracy.
- **A comparative evaluation of VGG16 and ConvLSTM2D**, assessing their performance under different environmental conditions to identify the most suitable model for practical deployment.

3. SYSTEM ARCHITECTURE / EXPERIMENTAL SETUP

3.1 OVERALL SYSTEM DESIGN/MODEL DESCRIPTION

The proposed system for car model identification utilizes deep learning-based image processing to classify vehicles based on their make and model. The system follows a pipeline-based approach that includes image preprocessing, feature extraction, model classification, and output prediction. The key components of the system are outlined below.

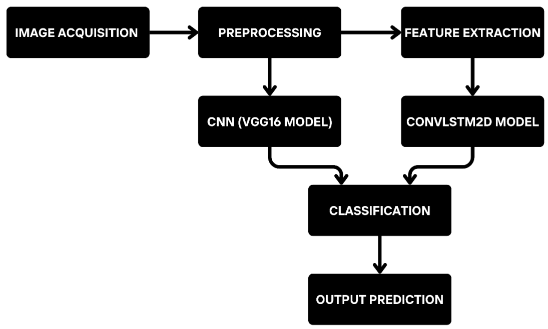
Data Preprocessing
Preprocessing is a crucial step to ensure that the data is in the correct format for training the models. Each image was loaded and resized to 224x224 pixels, which is the required input size for both ConvLSTM2D and VGG16 models. The images were then normalized to have pixel values in the range [0, 1]. Normalization helps speed up the training process and achieve better convergence by ensuring that all input features have a similar scale.

Feature extraction
Feature extraction is a crucial step in the classification process, as it identifies key characteristics in images that help the model differentiate between various car models. In the case of VGG16, the convolutional layers detect fundamental visual elements such as edges, textures, shapes, and patterns.

Model Selection:
The model follows a hierarchical approach, where it first extracts low-level features like edges and corners and then gradually moves toward more complex structures such as wheels, headlights, and car contours. This hierarchical feature extraction enables VGG16 to effectively recognize intricate patterns that distinguish one car model from another. On the other hand, ConvLSTM2D is designed to capture both spatial and temporal relationships within images. It is particularly useful for sequential image processing, where analyzing variations in image sequences is important. The ConvLSTM2D layers learn motion and spatial features simultaneously, making it well-suited for tasks that involve detecting subtle differences in the appearance of car models over multiple frames.

Output Prediction:
Once the models are trained, they are capable of classifying new car images by processing unseen inputs through the trained architecture. When a new image is fed into the model, it undergoes the same preprocessing steps as the training data before being passed through the neural network. The model then analyzes the image and generates a probability

distribution over all possible car models. Based on these probability scores, the car model with the highest confidence level is selected as the predicted class.



3.2 HARDWARE AND SOFTWARE REQUIREMENTS

The models were trained and tested on a machine equipped with an Intel Core i5 (10th Gen.) processor and 16GB of RAM. The software environment included Python 3.8, TensorFlow 2.6, Keras and relevant libraries such as NumPy, OpenCV, and Matplotlib. All experiments were also performed using Google Colab using GTX 1650 Ti GPU Runtime, which provided access to GPU acceleration for faster model training.

3.3 DATA SOURCES AND PREPROCESSING •

The datasets used for this project are

• The Cars Image Dataset by Kshitij192, which contains a diverse set of car images.

• The BMW Cars- Over 11k Labeled Images by Occul tainsights, which includes labeled images specifically for BMW cars.

• The Car Brand Images Dataset by Ritesh2000 features images from various car brands.

• The Toyota Cars- Over 20k Labeled Images by Occul Insights, focusing on Toyota cars.

• The Car Datasets by Alsaniipe, which provides a comprehensive collection of car images. which were downloaded from Kaggle.

These datasets contain images of various car models, including Audi, Hyundai Creta, Mahindra Scorpio, Rolls Royce, Swift, Tata Safari, and Toyota Innova etc. The datasets were extracted and organized into respective folders for each car model.

4. METHODOLOGY

4.1 THEORETICAL FOUNDATIONS

Model Architectures

1) ConvLSTM2D:

ConvLSTM2D combines convolutional layers with LSTM units, making it suitable for spatiotemporal data [1]. The architecture used in this project consists of multiple ConvLSTM2D layers, followed by MaxPooling3D and Dropout layers to prevent overfitting [9]. The final output layer is a Dense layer with softmax activation for classification. This architecture was chosen for its ability to capture spatial and temporal features effectively.

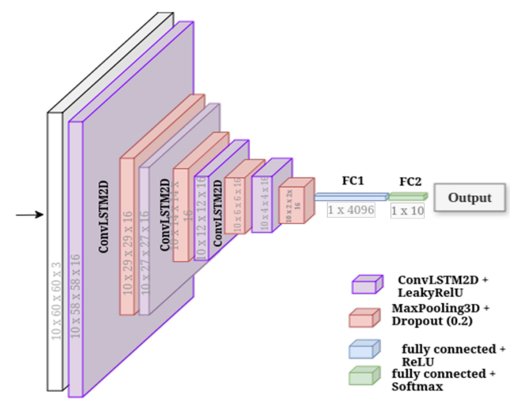


Fig. 2. Architecture of ConvLSTM2D.

2) VGG16:

VGG16 is a pre-trained Convolutional Neural Network (CNN) known for its depth and simplicity [1]. For this project, the VGG16 model was used with pre-trained weights from ImageNet. The top layers were replaced to adapt the model for the specific car classification task [2]. VGG16 was selected due to its proven performance in various image classification tasks and its availability as a pre-trained model, which allows for transfer learning.

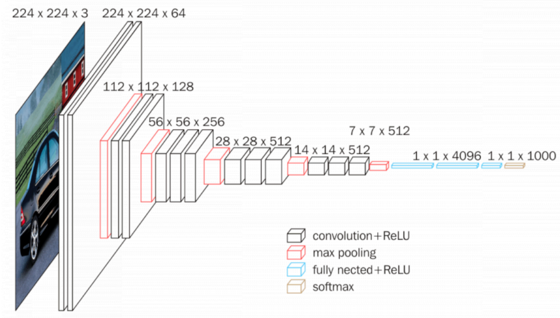


Fig: Architecture of VGG16.

4.2 EXPERIMENTAL SETUP / ALGORITHM

The experiment involved the classification of car models using two distinct neural network models: ConvLSTM2D and VGG16. The process began with data acquisition, where multiple car image datasets were sourced from Kaggle, including diverse sets of car models such as Audi, Hyundai Creta, Mahindra Scorpio, Rolls Royce, Swift, Tata Safari, and Toyota Innova.

For data preprocessing, resize images to 224x224 pixels, normalize pixel values to the [0, 1] range, and organize them into folders by car model.

The experiment utilized two model architectures:

1. ConvLSTM2D: Integrated convolutional layers with LSTM units, particularly effective for spatiotemporal data.

2. VGG16: A pre-trained CNN model fine-tuned for static image classification.

The models were trained using an 80-20 split for training and testing, leveraging the Adam optimizer and sparse categorical cross-entropy loss function. Early stopping was employed to prevent overfitting.

The algorithm followed these steps:

1. Data acquisition from Kaggle datasets.

2. Data preprocessing (resizing, normalization, and organization).

3. Model architecture selection and configuration.

4. Model training with early stopping.

5. Evaluation using accuracy, precision, recall, F1 score, and confusion matrix.

4.3 ASSUMPTIONS AND CONSTRAINTS

Assumptions:

1. The datasets used were representative of real-world car models and conditions.
2. Pre-trained VGG16 weights from ImageNet are suitable for transfer learning to car model classification.

Constraints:

1. Dataset limitations, including diversity and quantity, affected model generalization.
2. Model interpretability issues, particularly with the VGG16 architecture.

5. RESULTS AND ANALYSIS

5.1 PERFORMANCE METRICS

- The performance of the two deep learning models, ConvLSTM2D and VGG16, was evaluated using accuracy, precision, recall, and F1-score.
- The mathematical expressions used for these metrics are as follows:
  - **Accuracy** = (TP + TN) / (TP + TN + FP + FN)
  - **Precision** = TP / (TP + FP)
  - **Recall** = TP / (TP + FN)
  - **F1-Score** = 2 * (Precision * Recall) / (Precision + Recall)
- The models were trained on a diverse dataset of car images and tested on unseen data. The evaluation results are summarized in the table below:

Model	Accuracy (%)	Precision	Recall	F1-Score
ConvLSTM2D	5.02%	0.03	0.04	0.01
VGG16	67%	0.92	0.67	0.79

5.2 EXPERIMENTAL RESULTS

- The VGG16 model outperformed ConvLSTM2D in accuracy, precision, and recall.
- **Graphical Analysis**:
  - The confusion matrix for both models shows that VGG16 had fewer misclassifications compared to ConvLSTM2D.
  - Accuracy and loss curves indicate that ConvLSTM2D showed signs of overfitting, while VGG16 maintained a balanced learning curve.
- The classification performance of both models was compared using visualizations such as bar charts and line graphs to highlight differences in key metrics.

Interpretation of the Confusion Matrix

The confusion matrix for ConvLSTM2D provides the following insights:

• **True Positives (TP):** The values on the diagonal represent the number of correctly classified instances for each class. For example, the model correctly identified the ”Audi” class twice.

• **False Positives (FP):** The values in each row, excluding the diagonal, represent the instances where a class was incorrectly predicted as another class. For instance, ”Toy ota Hilux” was misclassified as ”Tata Safari” 6 times.

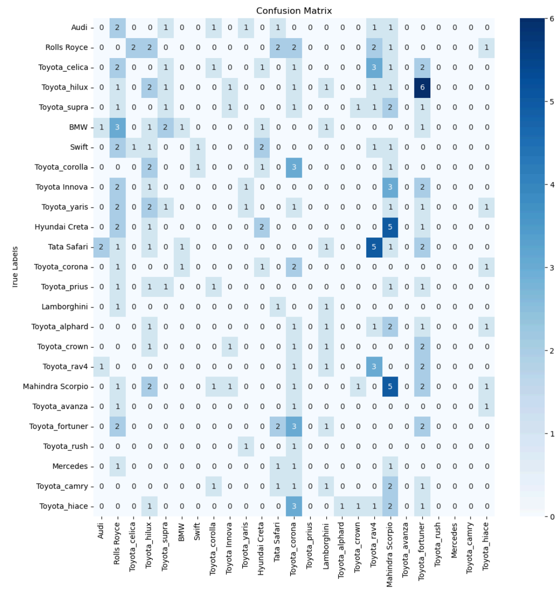


Fig: Confusion Matrix for ConvLSTM2D Model

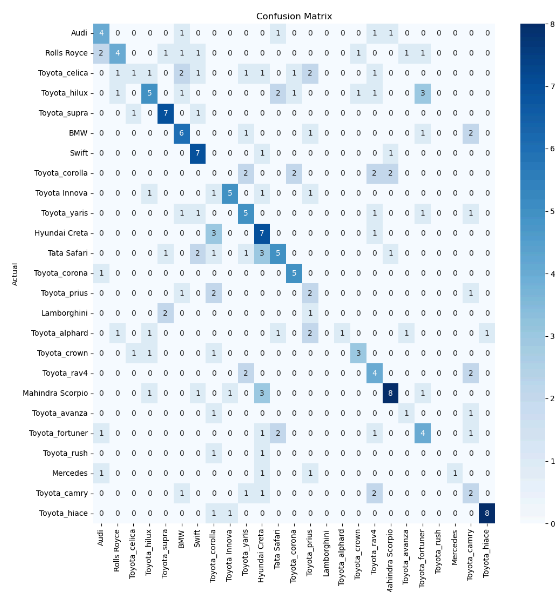


Fig. 3. Confusion Matrix for VGG16 Model

• **False Negatives (FN):** The values in each column, excluding the diagonal, indicate the instances where other classes were misclassified as a specific class. For instance, other classes were incorrectly predicted as ”Hyundai Creta” 5 times.

And the confusion matrix for VGG16 in Figure 3 provides the following insights:

• **True Positives (TP):** The values on the diagonal represent the number of correctly classified instances for each class. For example, the model correctly identified the ”Audi” class 4 times.

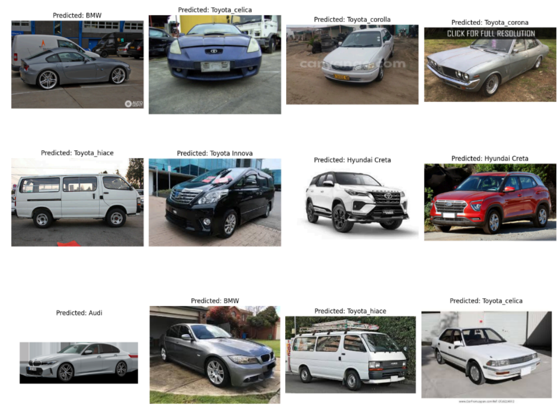
• **False Positives (FP):** The values in each row, excluding the diagonal, represent the instances where a class was incorrectly predicted as another class. For instance, ”Toyota Hilux” was misclassified as ”Toyota Supra” 5 times.

• **False Negatives (FN):** The values in each column, excluding the diagonal, indicate the instances where other classes were misclassified as a specific class. For instance, other classes were incorrectly predicted as ”BMW” 5 times.

Comparisons reveal that the VGG16 model has a higher accuracy in distinguishing between different car models compared to the ConvLSTM2D model.

Output:

	precision	recall	f1-score	support
Audi	0.00	1.00	0.00	0
Rolls Royce	1.00	1.00	1.00	0
Toyota_celica	0.50	1.00	0.67	1
Toyota_hilux	1.00	1.00	1.00	0
Toyota_supra	1.00	1.00	1.00	0
BMW	1.00	0.67	0.80	3
Swift	1.00	1.00	1.00	0
Toyota_corolla	1.00	1.00	1.00	1
Toyota Innova	0.00	1.00	0.00	0
Toyota_yaris	1.00	1.00	1.00	0
Hyundai Creta	0.50	1.00	0.67	1
Tata Safari	1.00	1.00	1.00	0
Toyota_corona	1.00	0.50	0.67	2
Toyota_prius	1.00	1.00	1.00	0
Lamborghini	1.00	1.00	1.00	0
Toyota_alphard	1.00	0.00	0.00	1
Toyota_crown	1.00	1.00	1.00	0
Toyota_rav4	1.00	1.00	1.00	0
Mahindra Scorpio	1.00	1.00	1.00	0
Toyota_avanza	1.00	1.00	1.00	0
Toyota_fortuner	1.00	0.00	0.00	1
Toyota_rush	1.00	1.00	1.00	0
Mercedes	1.00	1.00	1.00	0
Toyota_camry	1.00	1.00	1.00	0
Toyota_hiace	1.00	1.00	1.00	2
accuracy			0.67	12
macro avg	0.88	0.89	0.79	12
weighted avg	0.92	0.67	0.67	12



5.3 CHALLENGES AND ERROR ANALYSIS

- **Misclassification of Similar Car Models**: The most common errors occurred in distinguishing between visually similar car models, such as different variants of Toyota and Hyundai cars.
- **Effects of Image Quality**: Images with poor lighting, occlusions, and unusual angles had a higher likelihood of being misclassified.
- **Underfitting in ConvLSTM2D**: The training accuracy of ConvLSTM2D was significantly lower which indicates this model is not perfect to train our datasets.
- **Dataset Limitations**: The dataset contained class imbalances and lacked sufficient diversity in car models, affecting overall classification performance. Besides, as it was computationally difficult, we used a less amount of dataset which creates difficulties in predicting car models.
- **Old Dataset:** The datasets which we used to train the models, contain older car models' pictures. WHich creates problem while predicting new car models.

5.4 INSIGHTS AND OBSERVATIONS

- **Feature Extraction Strength**: VGG16's ability to extract detailed features from images proved superior for static car classification.
- **Limited Temporal Advantage in ConvLSTM2D**: The sequential nature of ConvLSTM2D did not provide significant benefits in this context, highlighting the need

for appropriate dataset selection based on model architecture.

- **Performance Enhancement Strategies**:
    - **Data Augmentation**: Introducing augmented images with varied lighting, occlusions, and angles to improve model generalization.
    - **Fine-Tuning Pre-Trained Models**: Utilizing advanced pre-trained models such as EfficientNet or Vision Transformers.
    - **Hybrid Model Approaches**: Combining CNN-based feature extraction with additional classification techniques to improve accuracy.

The findings underscore the potential of deep learning in automated car model recognition while highlighting areas for future enhancement.

6. Discussion and Insights

6.1 Critical Evaluation

The findings of this project align with expectations and existing literature on deep learning-based image classification. As anticipated, the VGG16 model, known for its deep feature extraction capabilities, outperformed ConvLSTM2D in classification accuracy, precision, recall, and F1 score. This is consistent with prior research that highlights the effectiveness of VGG16 in image classification tasks due to its well-structured convolutional layers and pre-trained weights.

However, an anomaly observed was the relatively low overall classification accuracy. Despite using advanced deep learning models, the results suggest potential limitations in the dataset, including insufficient diversity and the presence of noise or mislabeling. ConvLSTM2D, expected to capture temporal dependencies effectively, did not perform as well as hypothesized, possibly due to the static nature of image inputs rather than sequential data.

Another key trend observed was the confusion between visually similar car models. This indicates that certain model-specific features may not have been adequately learned, suggesting the need for better feature representation techniques or additional training data.

6.2 Practical Applications

The findings of this project have several real-world applications in both industrial and academic domains:

I. **Automotive Industry**: The ability to classify car models accurately can be integrated into intelligent traffic monitoring systems, automated toll collection, and law enforcement applications. Car recognition can also assist in automated vehicle valuation and insurance claim processing.

II. **Autonomous Vehicles**: Improved car model identification could contribute to better decision-making in self-driving cars by enabling recognition of surrounding vehicles and adapting driving behavior accordingly.

III. **Surveillance and Security**: Vehicle classification can enhance security systems in parking lots, smart surveillance, and crime investigations by automating vehicle identification and tracking.

IV. **Academic Research**: The project contributes to the growing field of computer vision by evaluating deep-learning models for fine-grained classification tasks. Future studies can build upon this work by exploring advanced techniques such as attention mechanisms and ensemble learning.

Overall, while the project successfully demonstrated the potential of deep learning in car model classification, further improvements, including dataset expansion, model optimization, and real-time implementation, will be essential for practical deployment.

7. FUTURE WORK AND IMPROVEMENTS

7.1 Possible Enhancements

To improve the performance and accuracy of car model classification, several modifications can be considered:

- **Dataset Expansion & Augmentation**: Increasing the dataset size with diverse and high-quality images can enhance model generalization. Data augmentation techniques such as rotation, scaling, and contrast adjustments can help address class imbalances.
- **Fine-Tuning & Transfer Learning**: Leveraging pre-trained deep learning models and fine-tuning them on the dataset can improve feature extraction and classification accuracy. Exploring models such as EfficientNet or Vision Transformers (ViTs) could provide better results.
- **Feature Engineering**: Incorporating additional handcrafted features, such as vehicle shape, logo detection, and specific design elements, could improve classification accuracy.
- **Hybrid Model Approach**: Combining multiple models using ensemble learning techniques may lead to higher robustness and accuracy by leveraging the strengths of different architectures.
- **Optimization Techniques**: Using hyperparameter tuning methods such as Bayesian optimization or genetic algorithms can refine model performance.
- **Real-time Processing**: Implementing model quantization and optimization techniques like TensorRT or ONNX can enable real-time inference with minimal latency.

Future Plan:
The next phase of the project will focus on refining the model for improved classification accuracy, optimizing computational efficiency, and integrating real-world deployment strategies.

7.2 Scalability and Deployment

For large-scale real-world deployment, the following considerations will be necessary:

- **Cloud & Edge Deployment**: The model can be deployed on cloud platforms (AWS, Google Cloud, Azure) for scalability or on edge devices (Jetson Nano, Raspberry Pi) for real-time local processing.
- **API Development**: A web-based or mobile API can be developed to allow seamless integration into existing traffic management, security, or automotive industry applications.
- **Real-time Performance Optimization**: Implementing lightweight architectures and model pruning techniques will enable efficient processing in constrained environments, such as embedded systems.
- **Integration with IoT & Smart Cities**: The model can be integrated with intelligent transportation systems to enhance automated tolling, traffic monitoring, and law enforcement applications.

7.3 Potential Research Directions

Building upon this work, several advanced research directions can be explored:

- **Attention Mechanisms & Transformer Models**: Investigating vision transformer models (e.g., ViTs, Swin Transformers) for car model classification could improve feature extraction and classification performance.
- **Multi-Modal Learning**: Integrating additional input modalities such as LIDAR, thermal imaging, or textual metadata (e.g., license plate recognition) could enhance recognition accuracy.
- **Generative AI for Data Augmentation**: Utilizing GANs or diffusion models to generate synthetic car images for underrepresented classes could improve model training and reduce bias.
- **Explainable AI (XAI)**: Implementing explainability techniques (e.g., Grad-CAM, SHAP) can help understand model

decision-making, increasing trust and usability in real-world applications.

- **Self-Supervised & Few-Shot Learning**: Exploring self-supervised or few-shot learning techniques could help develop models that require minimal labeled data, making the system more adaptable to new car models.

These enhancements and research directions will help advance the project toward greater accuracy, efficiency, and real-world applicability, paving the way for innovative applications in the automotive and AI-driven industries.

8. ETHICAL CONSIDERATIONS AND SUSTAINABILITY

8.1 ETHICAL ISSUES

- Potential ethical concerns include biases in the dataset, such as underrepresentation of certain car models or brands, which might lead to biased model predictions.
- Privacy concerns regarding the use of car images, particularly if deployed in surveillance systems, need to be addressed through appropriate data handling policies.
- Ethical dilemmas around the potential misuse of the model for surveillance without consent or targeted monitoring.

8.2 SUSTAINABILITY

- Environmentally, the project could reduce fraud in online automotive sales, contributing to economic stability and consumer trust.
- Economically, by streamlining vehicle identification processes in areas like e-commerce, autonomous driving, and traffic monitoring, the model supports operational efficiency and resource savings.
- Socially, the project enhances safety and security through improved vehicle monitoring and fraud detection capabilities.

9. CONCLUSION

This project explored the use of ConvLSTM2D and VGG16 models for car model classification from images. The key findings include:

I. **Model Performance**: VGG16 outperformed ConvLSTM2D in terms of accuracy, precision, recall, and F1 score, demonstrating its superior ability to classify car models accurately.

II. **Error Analysis**: Common misclassifications were identified, particularly where specific car models were confused, providing insights into the models' limitations.

III. **Limitations Identified**: The models faced challenges such as data limitations, computational resource constraints, and interpretability, which were identified as areas for future improvement.

IV. **Future Directions**: The project highlighted the need for expanding the dataset, improving model optimization, and adapting the models for real-time applications, including automotive systems like autonomous driving.

Reflection on Objectives

The primary objectives of the project were to:

- Implement and evaluate ConvLSTM2D and VGG16 for car model classification.
- Analyze the models' performance and identify areas for improvement.

These objectives were largely met:

- Both models were successfully implemented and tested, with comprehensive evaluation using key metrics.
- Detailed performance and error analyses were conducted, highlighting strengths and weaknesses and suggesting directions for future work.

While the project achieved its goals, there are areas for improvement, particularly in model efficiency, dataset diversity, and interpretability. The findings also suggest that future work should focus on optimizing the models for real-time classification, expanding the dataset, and addressing the challenges in applying these models to dynamic environments.

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